EE180 Fall Homework 1

Due: October 8, 2026, 23:59

Name: _____ **ID:** _____

Requirements:

- Use MATLAB or Python; Python is recommended.
- Submit a ZIP archive named `EE180-ID-Name-hw1.zip`, containing:
  - `codes/` folder with your source code,
  - `image/` folder with generated figures,
  - `report-ID-Name-hw1.pdf`,
  - `predictions_stage1.csv` and `predictions.csv` for the two stages of Question 3.
- Use relative file paths and include brief instructions for running your code.
- Plot all time–frequency representations using magnitude in decibels:

$$M_{\mathrm{dB}}(t, f) = 20 \log_{10} |X(t, f)|,$$

  where $X(t, f)$ denotes the normalized transform coefficients. Use the same normalization reference and color scale for plots compared within the same transform. For CWT, use the corresponding coefficient magnitude and map scales to frequency. State the normalization used, and apply a small positive floor before taking the logarithm to avoid $\log_{10}(0)$.
- Write your report in English.

1. Fourier and Time–Frequency Analysis

Use **Signal A** and **Signal B** in `data/q1_signals.mat` or `data/q1_signals.npz`. Both files contain `signal_A`, `signal_B`, `time_s` (seconds), and `sampling_rate_hz`. The sampling frequency is 60 MHz. Use μs and MHz for your time and frequency axes.

(a) **Time-domain and FFT analysis.**

Plot both signals in the time domain and their single-sided FFT magnitude spectra, using the same axis limits for corresponding plots and the same FFT normalization. Describe the observed differences and explain why the FFT magnitude spectra alone do not clearly distinguish their temporal structures.

(b) **STFT analysis.**

Plot the STFT magnitude spectrogram of each signal using a Hann window of 240 samples, a hop size of 6 samples, and $N_{\mathrm{FFT}} = 4096$. Plot the magnitude in dB as specified in the Requirements, using a common normalization reference and color scale for both signals. State how signal boundaries are handled.

Describe the differences between the two signals revealed by the STFT, supporting your interpretation with evidence from the plots.

Submission: Include your plots, explanations, and reproducible code.

2. STFT, CWT, and Time–Frequency Resolution

Use **Signal A** from Question 1 to investigate the effect of window length on STFT analysis and compare the results with continuous wavelet analysis.

(a) **STFT analysis and signal components.**

Choose a short window and a long window, and compute and plot the STFT magnitude spectrogram of Signal A for each. State and justify your parameter choices. Plot the magnitude in dB as specified in the Requirements. Use the same window type, magnitude normalization reference, and color scale for the short- and long-window results.

Hint: You may explore several window lengths, but only two representative results, one short and one long, are required.

Use the STFT results together to describe the distinct signal components present in Signal A. Estimate their characteristic frequencies, occurrence times, and frequency evolution where applicable. For components that are close in time or frequency, estimate their separation and indicate which spectrogram supports each estimate.

Support your interpretation with annotations, enlarged time–frequency regions, or one-dimensional slices through the STFT magnitude.

(b) **Time–frequency resolution trade-off.**

Using examples from your results, explain the trade-off between time and frequency resolution as the window length changes. Discuss whether one window length can adequately resolve all components of Signal A.

(c) **Continuous wavelet analysis.**

Analyze Signal A using the continuous wavelet transform with a complex Morlet wavelet, and plot its magnitude in dB against time and frequency, following the Requirements. State the wavelet parameters, scale or frequency range, and coefficient normalization used.

Compare your result with the STFT results in part (a). Discuss how well the different signal components can be distinguished, and explain your observations in terms of the time and frequency localization of the STFT window and the wavelet. Does CWT eliminate the time–frequency resolution trade-off? Support your answer with evidence from your results.

Hint: You may explore different wavelet parameters, but presenting one representative CWT result is sufficient.

Submission: Include your STFT and CWT plots, parameter choices, component identification, discussion, and reproducible code.

3. Classifying Classroom Ultrasound Recordings

In class, you will acquire ultrasound signals from three objects or sample configurations following the TA's demonstration. Each student will make three independent acquisitions per class. Differences in handling may cause recordings of the same class to vary.

Each supplied recording is a **one-dimensional signal with 896 samples**. The **sampling frequency is** 26.5216 MHz. The supplied signals have already been normalized to the range $[-1, 1]$; no raw-ADC normalization is required.

You will receive:

- A labeled reference set collected by the TA: **15 independent recordings per class** (45 in total).
- **Stage 1:** An additional set collected by the TA, with **5 recordings per class** (15 in total), separate from the reference set, anonymized and shuffled, with class labels withheld.
- **Stage 2:** The classroom recordings, anonymized and shuffled, with class labels withheld.

Use signal analysis methods introduced in class to extract features from the ultrasound recordings and classify each recording in both stages as **Class 1, 2, or 3**. Submit **one label per recording**. A README in the dataset folder will describe the file formats, identifiers, data layout, units, and acquisition metadata.

Signal decomposition and feature extraction.

Apply appropriate signal decomposition or transformation methods to analyze the reference recordings. Represent each recording by one set of features. Compute descriptive statistics of the feature values for each class, examine differences between classes and variation within each class, and support your observations with plots or summary tables. Apply the same preprocessing and feature-extraction procedure to the reference set and both task stages. If you include time–frequency plots, follow the dB display convention in the Requirements.

Classification.

Use the feature statistics and plots from the labeled reference set to **manually choose thresholds or characteristic ranges** for distinguishing the three classes. State your decision rules as explicit numerical conditions, explain how you chose them, and apply the same rules in both stages.

As an **optional alternative**, you may form a feature vector for each recording and train a machine-learning classifier using the labeled reference set. **Machine learning is not required.**

Describe your features, decision rules or optional model, and parameter choices. Use the labeled reference set to determine your thresholds or model parameters. Base predictions on the recorded signals rather than acquisition order or knowledge of which object was measured.

Two-stage task.

In **Stage 1**, classify the 15 additional TA recordings. In **Stage 2**, classify each shuffled classroom recording. Labels are withheld in both stages; submit one predicted label per recording. The TA will evaluate both submissions against private ground-truth labels. Students are not required to report task-stage accuracy or analyze errors against withheld labels.

Submission and assessment: Both stages must be submitted and are included in the assessment. Submit `predictions_stage1.csv` for Stage 1 and `predictions.csv` for Stage 2. Each file must contain the columns `sample_id` (the recording identifier) and `predicted_class`, with one row per recording in that stage. Encode `predicted_class` as 1, 2, or 3. Describe your method and reference-set analysis in the report, including supporting figures, and provide reproducible code.